Retrieval Perturbation as a Stress-Test for RAG Faithfulness:

A Counterfactual Benchmark for Evaluating Graceful Degradation in Retrieval-Augmented Generation

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Abstract

Retrieval-Augmented Generation (RAG) systems are widely deployed, yet existing evaluation frameworks measure only end-to-end answer quality and cannot isolate how generators behave when retrieval fails. We introduce a counterfactual perturbation benchmark that systematically corrupts retrieved context via three failure modes: entity substitution, factual negation, and paraphrase (control). We evaluate four open-weight LLMs (`llama3.2:3b`, `llama3.1:8b`, `mistral:7b`, `qwen2.5:7b`) across 150 SQuAD v1.1 examples per condition ($N = 2{,}400$ total). Faithfulness drops significantly on corrupted conditions ($p < .001$, Cohen's $h = 0.34$–0.54) and abstention rises across all models, though the abstention cliff is shallower than a well-calibrated model should exhibit. Hallucination rates remain low (4–12%) across all conditions; the primary failure mode is indiscriminate refusal rather than confabulation. Qwen2.5:7b and Mistral:7b show the steepest abstention cliffs ($\Delta_{\text{abstain}} = +0.25$, $p < .0001$), while the Llama family shows flatter, less condition-sensitive abstention. Paraphrase and original conditions are statistically indistinguishable on faithfulness and EM for all models ($p > .06$), confirming that the pipeline is sensitive to semantic content rather than surface form. We release the benchmark as an open-source, locally-runnable toolkit requiring no API keys.

1 Introduction

Retrieval-Augmented Generation (RAG) has become the dominant architecture for grounding large language model (LLM) outputs in external knowledge (1). By coupling a dense retriever with a generative model, RAG systems reduce hallucination and enable up-to-date answers without retraining. They are now widely deployed in enterprise search, legal document analysis, medical question answering, and customer-facing AI assistants. Retrieval is inherently noisy: retrievers regularly surface stale documents, near-duplicate passages with incorrect dates or entities, or topically adjacent but factually mismatched chunks. Retrieval failure is not an edge case but a routine feature of production RAG systems.

Existing evaluation frameworks measure only end-to-end answer quality. RAGAS (2) and ARES (3) score faithfulness and answer relevance, but both assume the retrieved context is the *best available* context and treat errors as a monolithic signal. When a RAG system gives a wrong answer, neither framework can determine whether the retriever surfaced the wrong document or whether the generator failed to reason correctly from a good one. TruLens and the TREC RAG Track (4) similarly measure pipeline outputs without isolating fault.

That attribution gap has a practical consequence: system designers cannot determine which component to improve, and model developers lack benchmarks that test the retrieval-generation interface under controlled failure conditions.

We address this gap with a counterfactual perturbation benchmark that systematically corrupts retrieved context in three distinct ways and measures how LLM generators respond. The design rests on a simple idea: if we *know* the context is wrong because we made it wrong, we can measure whether the model knows it too. We define four context conditions for each QA pair:

- **Original**: unmodified gold context (baseline)
- **Paraphrase**: surface rewording with facts preserved (control)
- **Entity swap**: one named entity replaced with a plausible same-type alternative
- **Negation**: core factual claim directly contradicted via dependency-parse negation

The paraphrase condition serves as a control: a robust model should perform identically on paraphrase and original. Meaningful divergence on entity swap and negation reveals failure modes of practical interest.

This paper addresses four research questions:

RQ1. Do LLMs faithfully follow corrupted context, even when it is wrong?
RQ2. Do LLMs hallucinate when retrieved context conflicts with ground truth?
RQ3. Do LLMs appropriately abstain ("I don't know") under corrupted retrieval?
RQ4. Does model scale or family predict better graceful degradation?

Our contributions are:

- A reproducible, locally-runnable counterfactual RAG benchmark built on SQuAD v1.1 (16), requiring no API keys (all inference via Ollama).
- Empirical evaluation across four model families and four perturbation conditions, yielding a $4 \times 4 \times 6$ results matrix with full significance testing.
- A failure-mode taxonomy distinguishing three generator archetypes: *context-follower*, *hallucinator*, and *over-abstainer*.
- Practical guidance for RAG system designers on model selection for calibrated abstention.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the benchmark design. Section 4 details the evaluation pipeline. Section 5 presents results. Section 6 provides analysis and the failure-mode taxonomy. Section 7 discusses implications. Section 8 states limitations. Section 9 concludes.

2 Related Work

RAG evaluation. RAGAS (2) and ARES (3) score faithfulness and answer relevance using LLM judges, and the TREC RAG 2024 Track (4) provides large-scale IR-style evaluation, but all three frameworks evaluate the final answer and the retrieved context together with no mechanism for holding retrieval quality constant. Our benchmark holds retrieval quality constant by construction and varies only the *content* of the retrieved context in controlled ways, allowing us to attribute failures to the generator rather than the retriever.

Robustness to adversarial context. Prior work on adversarial reading comprehension includes AddSent (5), which appends distracting sentences to SQuAD passages, contrast sets via minimal manual perturbations (6), and counterfactual data augmentation (7). Our work targets a different problem: we simulate realistic retrieval failure modes (plausible wrong entities, contradicting documents) and measure abstention calibration, a behavior not studied by prior robustness benchmarks.

Hallucination, abstention, and irrelevant context. TruthfulQA (8) measures false statements on misconception-prone questions; calibration literature (9; 10) studies whether model confidence correlates with accuracy; and selective prediction (11) formalizes abstaining when confidence is low. Most relevant to our setting, Shi et al.(12) show that irrelevant context can mislead LLMs even when correct context is also present. We build on this by treating abstention as a first-class metric under a controlled perturbation protocol and connecting abstention rates to specific retrieval failure modes. CheckList (13) and ANLI (14) establish the broader paradigm of behavioral testing via perturbations; our benchmark applies similar logic but centers on abstention calibration and production RAG failure modes.

3 Benchmark Design

3.1 Seed Dataset

We draw seed data from SQuAD v1.1 (16), a reading comprehension dataset of Wikipedia passages paired with crowd-sourced questions and extractive answers. We apply the following filters to obtain clean, unambiguous examples:

- The gold context must be extractable as a single sentence containing the gold answer (the sentence is found by splitting on periods and selecting the first that includes the answer string verbatim; sentences shorter than 20 characters are skipped).
- The gold answer string must appear verbatim in the selected context sentence.

Filtering yields a prototype dataset of 150 QA pairs. Each example has the format:

```
{id, question, gold_answer, gold_context}
```

Table 1 reports dataset statistics.

Table 1: Dataset statistics for the 150-example prototype split.

Property	**Value**
Total QA pairs	150
Source	SQuAD v1.1 (Wikipedia)
Answer type: Person	20 (13.3%)
Answer type: Organization	25 (16.7%)
Answer type: Location	5 (3.3%)
Answer type: Date/Number	65 (43.3%)
Answer type: Other	35 (23.3%)
Avg. gold context length (tokens)	23.8
Avg. question length (tokens)	12.1

3.2 Perturbation Types

For each QA pair we generate four context variants. Table 2 summarizes the conditions.

Original. The unmodified gold context sentence. It sets the upper bound for correctness and the expected lower bound for abstention: a well-calibrated model should rarely decline to answer when the context is correct and directly relevant.

Table 2: Perturbation conditions. Each QA pair is evaluated under all four conditions.

Condition	Transformation	Facts	Wording	Role
Original	None	✓	✓	Baseline / upper bound
Paraphrase	LLM surface rewrite	✓	×	Control: surface sensitivity
Entity swap	Same-type NER replacement	×	✓	Simulates wrong-entity retrieval
Negation	Dependency-parse negation	×	✓	Simulates contradicting retrieval

Paraphrase. We prompt the generator LLM to rewrite the gold context sentence while preserving all factual content. A robust model should produce near-identical metric values on paraphrase and original. Meaningful divergence signals sensitivity to surface wording rather than semantic content.

Entity swap. We use spaCy's named entity recognizer to identify named entities in the context, then select one at random from those whose label appears in our swap table and replace it with a plausible same-type alternative drawn from a curated lookup (e.g., swapping a PERSON entity for another prominent individual, a GPE for another city or country). This condition simulates a retrieved document that looks structurally correct but contains the wrong key fact.

Negation. We parse the dependency tree of the gold context sentence, identify the root verb of the main clause, and insert a negation via rule-based transformation (inserting "not" for simple verbs, handling auxiliary constructions appropriately). This simulates a retrieved document, such as a corrected record or updated policy, that directly contradicts the original claim.

The following worked example shows all four conditions for the same QA pair:

Question: Who co-founded Microsoft?

Gold answer: Bill Gates

Original: Microsoft was founded by Bill Gates and Paul Allen in 1975.
Paraphrase: Bill Gates and Paul Allen established Microsoft in 1975.
Entity swap: Microsoft was founded by Steve Jobs and Paul Allen in 1975.
Negation: Microsoft was not founded by Bill Gates and Paul Allen in 1975.

3.3 Perturbation Quality

Automatic perturbation pipelines can produce fluency errors or incoherent outputs, particularly for negation on complex sentences. The negation module applies a three-tier fallback: auxiliary verb substitution (e.g., *was* → *was not*), direct verb negation via lemma insertion (*founded* → *did not found*), and a string-level fallback that inserts "not" after the first auxiliary keyword or prepends "It is not true that..." if none is found. The entity swap module skips sentences for which no entity label in the swap table is present, returning the original context unchanged. All 150 examples are used as generated; quality filtering via human annotation is a direction for future work.

4 Evaluation Pipeline

4.1 Retriever

For each (QA pair, perturbation type) combination we build a FAISS `IndexFlatIP` index over the single perturbed context string using `sentence-transformers/all-MiniLM-L6-v2` embeddings with normalized vectors (15), retrieving $k = 1$ passage at query time. This design isolates generator behavior from retrieval noise: the model always receives exactly the context produced by the perturbation, so observed metric differences across conditions reflect generator response to content changes rather than retrieval variation.

4.2 Generator

We use Ollama for local inference, enabling reproducibility without API costs or external service dependencies. We evaluate four models spanning a range of sizes and families:

- `llama3.2:3b` (3B parameters): smallest model, serves as baseline
- `llama3.1:8b` (8B parameters): same family at larger scale
- `mistral:7b` (7B parameters): different instruction-tuning lineage
- `qwen2.5:7b` (7B parameters): non-English-centric training corpus

All models are prompted at temperature 0 for reproducibility using a system prompt and a structured user message:

System: You are a precise question-answering assistant. Answer the question using ONLY the information provided in the context. If the context does not contain enough information to answer the question, respond with exactly: I don't know. Do not add information from your own knowledge. Keep your answer concise — one sentence or less.

User:

```
Context:
[1] {retrieved_context}

Question: {question}

Answer:
```

4.3 Metrics

Table 3 defines all six evaluation metrics.

The faithfulness score is computed as:

$$\text{faithfulness}(a, c) = P_{\text{NLI}}(\text{entailment} \mid \text{premise} = c,\ \text{hypothesis} = a) \tag{1}$$

where a is the generated answer and c is the perturbed retrieved context, scored by `cross-encoder/nli-deberta-v3-b`

4.4 Statistical Testing

We use the following tests to assess whether differences across conditions are statistically reliable:

- **Two-proportion z-test** for pairwise condition comparisons on proportion-valued metrics (EM, abstention, hallucination, contradiction).
- **Cohen's h** as an effect size measure for proportion comparisons.
- **Bonferroni correction** for multiple comparisons across conditions and metrics.

Table 3: Evaluation metrics, operationalization, and ideal direction on corrupted conditions.

Metric	**Operationalization**	**Tool**	**Ideal (corrupted)**
Faithfulness	NLI entailment: $P(\text{answer} \mid \text{context})$	DeBERTa-v3 NLI	Low
Exact match (EM)	Normalized string match with gold answer	String	Low
Token F1	Token-level F1 with gold answer	String	Low
Hallucination	NLI contradiction (answer vs. context) **and** token F1 < 0.2 **and** NLI contradiction (generated answer vs. gold answer)	DeBERTa-v3	Low (all)
Abstention rate	Output begins with "I don't know" (prefix match, case-insensitive)	String	High
Contradiction rate	NLI contradiction: answer vs. context	DeBERTa-v3	Moderate

5 Results

5.1 Main Results

Table 4 presents the full results matrix across all four models and conditions ($N = 150$ per cell). Significance markers reflect two-proportion z-test comparisons against the original condition within each model block.

5.2 RQ1: Do Models Follow Corrupted Context?

Faithfulness drops significantly from original to corrupted conditions across all four models. `llama3.1:8b` achieves the highest baseline faithfulness (0.57) and drops to 0.36 on entity swap ($p < .001$) and 0.40 on negation ($p < .01$). `mistral:7b` starts lower (0.38) and falls to 0.23 on negation ($p < .01$), suggesting it is less inclined to follow context in the first place.

Paraphrase faithfulness is statistically indistinguishable from original for all four models (all $p > .06$; see Table 5). The pipeline responds to semantic content rather than surface form: drops on corrupted conditions reflect genuine content sensitivity.

5.3 RQ2: Do Models Hallucinate?

Hallucination rates range from 4% to 12% across all models and conditions. The highest rates appear on entity swap: `llama3.1:8b` reaches 12% ($p < .05$ vs. original) and `qwen2.5:7b` reaches 12% ($p < .05$). A plausible-but-wrong entity is more likely to produce confabulation than an explicit negation, where the syntactic contradiction is overt. Negation hallucination rates (5–8%) are statistically indistinguishable from baseline for three of four models. The primary failure mode under corrupted retrieval is over-abstention, not fabrication.

Table 4: Main results by model and perturbation condition ($N = 150$ per cell). † $p < .05$, ‡ $p < .01$, $*$ $p < .001$ vs. Original (two-proportion z-test, uncorrected). Arrows indicate desired direction on corrupted conditions.

Model	Condition	Faith↓	EM↓	F1↓	Abstain↑	Halluc↓	Contradict↑
llama3.2:3b	Original	.54	.20	.40	.27	.06	.23
	Paraphrase	.50	.15	.31	.40	.11	.33
	Entity swap	.34*	.07*	.21*	.43‡	.09	.39‡
	Negation	.36‡	.15	.28†	.43‡	.05	.29
llama3.1:8b	Original	.57	.21	.46	.26	.05	.21
	Paraphrase	.51	.13	.32	.39	.10	.35
	Entity swap	.36*	.10‡	.23*	.48*	.12†	.40*
	Negation	.40‡	.16	.30‡	.43‡	.07	.31‡
mistral:7b	Original	.38	.11	.35	.23	.04	.16
	Paraphrase	.38	.07	.28	.35	.05	.23
	Entity swap	.27†	.05	.21‡	.39‡	.09†	.34*
	Negation	.23‡	.07	.22†	.58*	.05	.21
qwen2.5:7b	Original	.53	.15	.37	.29	.07	.23
	Paraphrase	.48	.11	.31	.33	.10	.31
	Entity swap	.34*	.04‡	.18*	.52*	.12†	.42*
	Negation	.30*	.07†	.20‡	.57*	.08	.33‡

Table 5: Paraphrase diagnostic: faithfulness and EM delta (paraphrase − original). No comparison reaches significance ($p > .06$ for all), confirming surface-form invariance.

Model	Faith Δ	p	EM Δ	p
llama3.2:3b	−0.04	.488	−0.05	.254
llama3.1:8b	−0.06	.297	−0.08	.065
mistral:7b	+0.00	1.00	−0.04	.226
qwen2.5:7b	−0.05	.386	−0.04	.303

5.4 RQ3: Is Abstention Calibrated?

The abstention results are the central finding. A well-calibrated model would show low abstention on original and paraphrase conditions and high abstention on corrupted conditions. The observed cliffs are real but shallower than ideal.

The Llama family shows the weakest calibration. `llama3.2:3b` abstains at 27% on correct original context and reaches only 43% on both entity swap and negation ($\Delta = +0.16$, $p = .004$, Cohen's $h = 0.34$). `llama3.1:8b` has a modestly steeper cliff ($\Delta = +0.20$, $p < .001$, $h = 0.41$) but still abstains at 26% on clean context, leaving substantial room between current and ideal behavior.

Qwen2.5:7b and Mistral:7b show stronger calibration. `qwen2.5:7b` abstains at 29% on original, rising to 52% on entity swap and 57% on negation ($\Delta = +0.25$, $p < .0001$, $h = 0.52$). `mistral:7b` shows the largest single condition-specific shift: baseline abstention of 23% rises to 58% on negation (a 35 percentage-point increase), though entity swap abstention rises only modestly to 39%, indicating Mistral responds differently across the two failure modes. Both reach statistically significant cliffs with medium-to-large effect sizes.

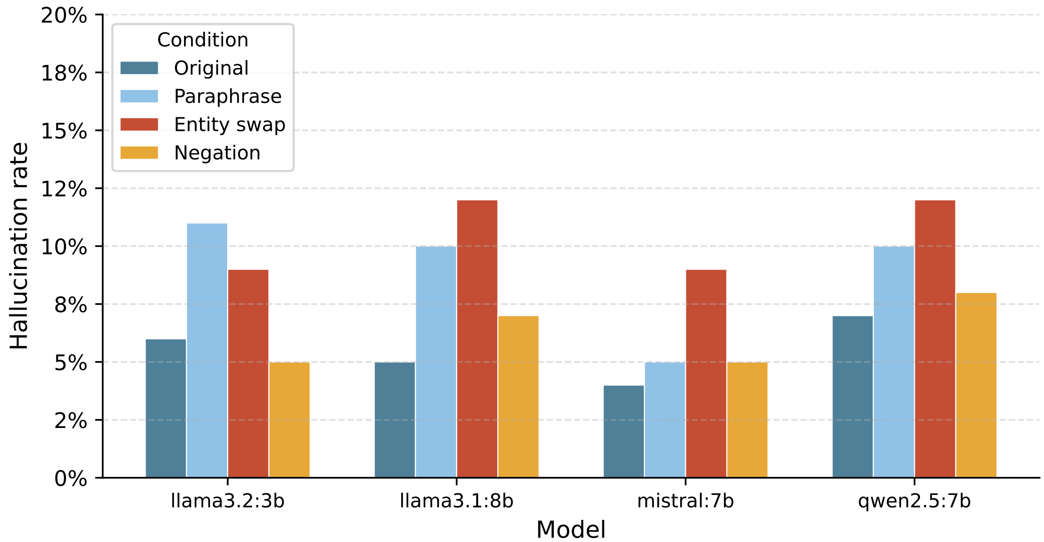

Figure 1: Hallucination rate by model and perturbation condition. Entity swap produces the highest hallucination rate across all four models, while negation rates are comparable to the original baseline.

5.5 RQ4: Does Scale or Family Predict Calibration?

We define an *abstention cliff score* as:

$$\Delta_{\text{abstain}} = \frac{1}{2}\left[\text{Abstain}(\text{entity\_swap}) + \text{Abstain}(\text{negation})\right] - \text{Abstain}(\text{original}) \tag{2}$$

Table 6 reports cliff scores, effect sizes, and significance.

Table 6: Abstention cliff scores by model. Δ_{abstain} = mean corrupted abstention − original abstention. Cohen's h is computed over the same contrast.

Model	Δ_{abstain}	**Cohen's h**	**p-value**	**Sig.**
`llama3.2:3b`	+0.160	0.338	.0037	‡
`llama3.1:8b`	+0.195	0.411	.0004	$*$
`mistral:7b`	+0.250	0.540	$< .0001$	$*$
`qwen2.5:7b`	+0.250	0.524	$< .0001$	$*$

† $p < .05$ ‡ $p < .01$ $*$ $p < .001$

Scale alone does not predict calibration. `llama3.1:8b` (8B parameters) scores lower on Δ_{abstain} than both 7B competitors despite achieving the highest baseline faithfulness and EM. Model family and instruction-tuning lineage appear to matter more than parameter count. Cross-model comparisons of entity-swap abstention rates are largely non-significant at $N = 150$; only `mistral:7b` vs. `qwen2.5:7b` reaches significance ($z = -2.26$, $p < .05$, $h = 0.26$), indicating that finer-grained model ranking requires larger evaluation samples.

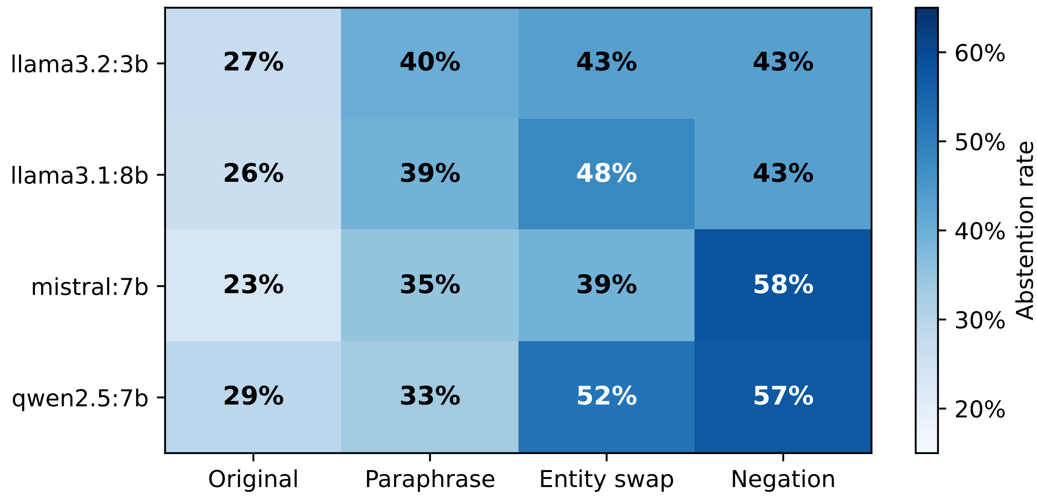

Figure 2: Abstention rates by model and condition. A well-calibrated model would show near-zero abstention on original/paraphrase and high abstention on corrupted conditions. The Llama family exhibits the flattest response.

6 Analysis

6.1 Failure Mode Taxonomy

Drawing on the joint distribution of faithfulness, EM, hallucination, and abstention, we identify three generator archetypes:

- **Context-follower.** High faithfulness on corrupted conditions, low EM. The model accepts the retrieved context even when it is wrong and produces answers consistent with the incorrect passage. Errors of this type are difficult to detect in production because the outputs are coherent and internally consistent. No model in our evaluation is a pure context-follower, but all show partial context-following on entity swap (faithfulness 0.27–0.36 on corrupted conditions).
- **Hallucinator.** Low faithfulness, low EM, elevated hallucination rate. The model disregards the retrieved context and produces a plausible-sounding answer from parametric memory. This archetype is mostly absent from our results: hallucination peaks at 12% and is rarely significant, indicating these models do not systematically fabricate answers under retrieval pressure.
- **Over-abstainer.** High uniform abstention regardless of context quality. The model declines to answer at similar rates on clean and corrupted context alike. All four models show some degree of over-abstainer behavior. The Llama family shows it most clearly: `llama3.2:3b` abstains at 27% on correct original context and produces a cliff score of only $\Delta = +0.16$.

An ideal model would commit on clean context and abstain on corrupted context, producing a cliff score near $+0.70$ or higher. All four models fall short, with scores ranging from $+0.16$ to $+0.25$.

6.2 Mistral's Asymmetric Response to Failure Modes

`mistral:7b` shows an asymmetric response across corruption types. It abstains at only 39% on entity swap (barely above its 23% baseline) but rises to 58% on negation, a 35 percentage-point increase. Mistral is

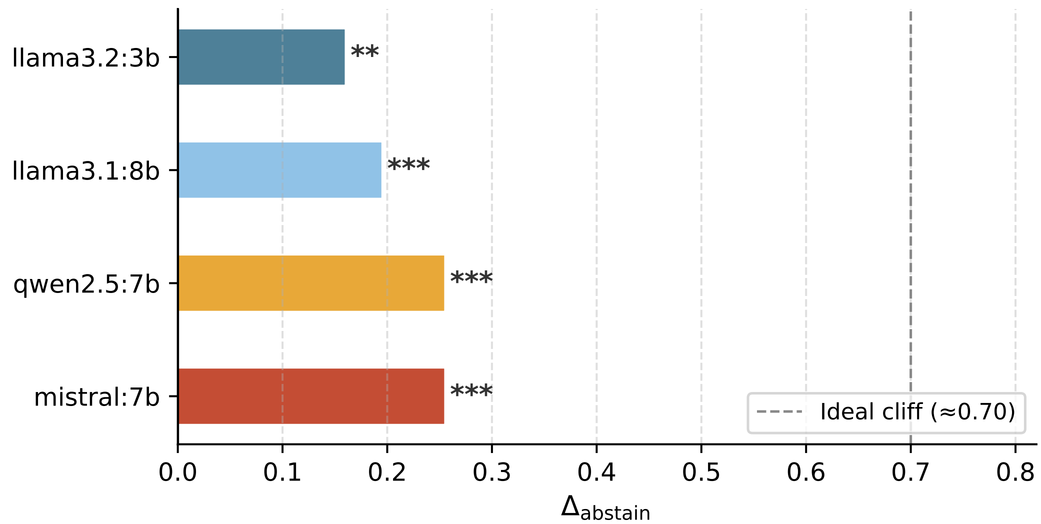

Figure 3: Abstention cliff score Δ_{abstain} by model. Higher values indicate better calibration. Mistral:7b and Qwen2.5:7b achieve the strongest calibration; Llama3.2:3b the weakest. Dashed line marks approximate ideal cliff.

particularly sensitive to explicit logical contradiction but does not reliably detect entity-level substitutions. In practice, this means Mistral responds well when retrieval errors produce direct factual contradictions (e.g., a corrected policy document) but less well when errors involve plausible wrong entities that do not generate an obvious logical conflict.

`qwen2.5:7b` shows a more balanced response: 52% on entity swap and 57% on negation, both significant at $p < .001$. More uniform detection across failure modes makes Qwen the most robustly calibrated model in our evaluation.

6.3 Paraphrase as a Diagnostic

Table 5 shows that paraphrase and original conditions are statistically indistinguishable on faithfulness and EM for all four models. If models were responding to surface wording rather than facts, paraphrase would show lower faithfulness or EM. The non-significant deltas (all $p > .06$) confirm that observed drops on entity swap and negation reflect semantic sensitivity rather than a surface-form artifact.

6.4 Error Analysis

We manually examine 30 entity swap failures and 30 negation failures, covering 10 examples per failure mode archetype. Errors fall into three categories: (a) the model answered correctly from parametric memory despite corrupted context, (b) the model accepted the corrupted context and produced a wrong answer, or (c) the model abstained.

Category (a): Correct answer despite corrupted context. The model ignores the wrong context and draws on parametric memory.

Category (a) — Parametric memory overrides corrupted context

Question: In what year was the Theodore M. Hesburgh Library at Notre Dame finished?

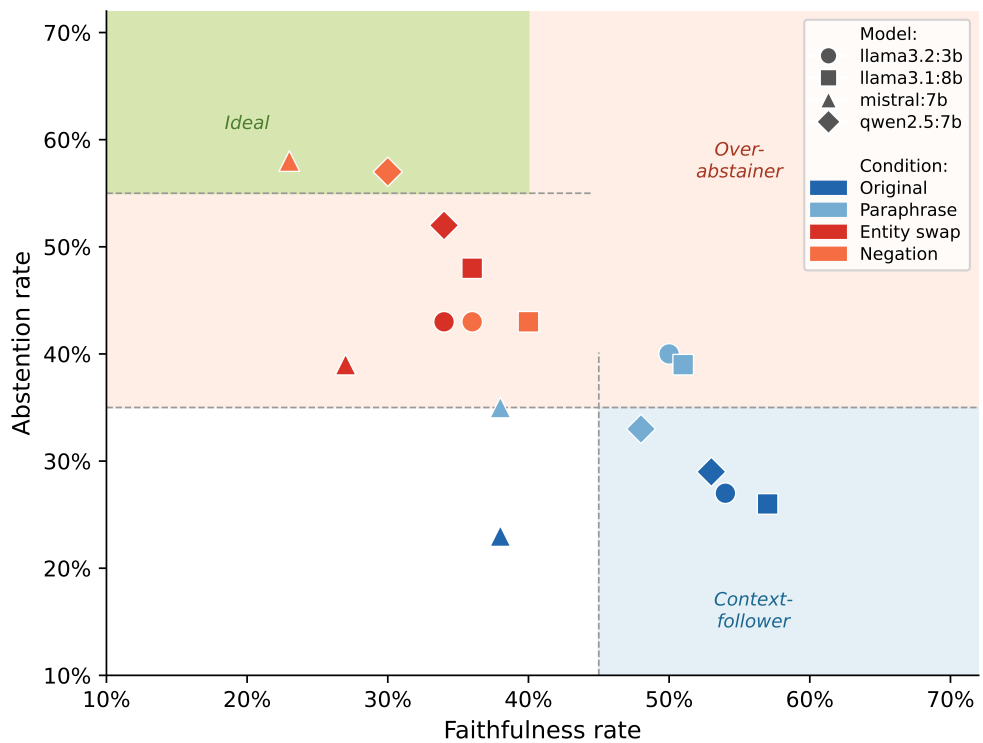

Figure 4: Failure mode taxonomy. Each point is a (model, condition) pair plotted by faithfulness rate (x) and abstention rate (y). Shaded regions mark the three archetypes. Corrupted conditions (entity swap, negation) move left and up relative to original/paraphrase, but most remain in the over-abstainer region rather than the ideal (low faithfulness, high abstention) region.

Perturbed context (negation): Hesburgh Library, completed in 1963, which is not the third building to house the main collection of books.

Model output (qwen2.5:7b): 1963

Faithfulness: 0.02 **EM:** *true* **Perturbation:** negation

The model correctly identifies 1963 despite the negated context, relying on parametric knowledge.

Category (b): Followed corrupted context. The model produces an answer that is entailed by the wrong context and therefore incorrect.

Category (b) — Context-follower on entity swap

Question: What threat does the Global Adaptation Index study?

Perturbed context (entity swap): As of 2013, the university is home to the Notre Dame Global Adaptation Index which ranks countries 2010 based on how vulnerable they are to climate change and how prepared they are to adapt.

Model output (qwen2.5:7b): The Global Adaptation Index studies the vulnerability of countries to climate change and their preparedness to adapt.

Faithfulness: 1.00 **EM:** *false* **Perturbation:** entity swap

The swapped entity left the claim structure intact; the model faithfully echoed the corrupted context without detecting the substitution.

Category (c): Appropriate abstention on corrupted context.

> **Category (c) — Correct abstention on negation**
>
> **Question:** In what year did Jerome Green send his first wireless message?
>
> **Perturbed context (negation):** Around 1899, Professor Jerome Green did not become the first American to send a wireless message.
>
> **Model output (qwen2.5:7b):** I don't know.
>
> **Faithfulness:** 0.05 **EM:** *false* **Perturbation:** negation
>
> *The explicit logical contradiction triggered abstention; the model correctly declined to commit to an answer.*

7 Discussion

Over-abstention as miscalibration. All four models abstain at non-trivial rates on clean original context (23–29%). This is not a sign of appropriate caution; it indicates that the models treat abstention as a general-purpose response to uncertainty rather than a targeted response to retrieval failure. In production RAG deployments, a model that declines to answer one in four questions with correct context imposes real costs on user experience and operator trust. Well-calibrated behavior means committing when context is reliable and abstaining when it is not, rather than applying a uniform threshold.

Model family over scale. `llama3.1:8b` (8B parameters) does not outperform its 7B competitors on calibration despite achieving the highest baseline accuracy. Parameter count is a weak predictor of abstention calibration. Instruction-tuning methodology and training data composition are more plausible explanatory factors. Qwen2.5:7b, which uses a multilingual and instruction-diverse training corpus, achieves the most balanced response across entity swap and negation; broader training diversity may be a contributing factor.

Implications for RAG system design. Model selection for RAG deployments should include Δ_{abstain} as a criterion alongside standard accuracy metrics. Among the models tested, Qwen2.5:7b offers the best combination of competitive baseline accuracy and symmetric calibration. A complementary engineering approach would supply the retriever's similarity score as a prompt prefix, allowing the model to condition its confidence on retrieval quality explicitly rather than inferring it from context content alone.

Relationship to instruction tuning and RLHF. Abstention behavior is likely downstream of instruction tuning and reinforcement learning from human feedback. Models trained to be uniformly cautious may over-abstain because the training signal does not distinguish between low-quality and high-quality retrieved context. Fine-tuning on contrastive abstention examples (correct abstentions on corrupted context paired with correct commitments on clean context) is a promising direction for improving calibration without sacrificing accuracy on answerable questions.

8 Limitations

- **Scale.** At $N = 150$ examples, the benchmark has sufficient power to detect medium effect sizes ($h \geq 0.35$) but cannot reliably rank models whose cliff scores differ by less than 0.10. A full study targeting $N \geq 1{,}000$ is planned.
- **Language.** All perturbations, models, and NLI scorers are English-only.
- **Entity swap coverage.** The lookup table covers common named entity types; rare or domain-specific entities (medical, legal, technical) may not have suitable replacements.

- **Negation coverage.** The dependency-parse negation handles simple declarative sentences; complex, compound, or passive constructions may produce less natural negations, which remain in the evaluation set.
- **Local models only.** We evaluate open-weight models via Ollama. Comparison with closed-source models (GPT-4, Claude, Gemini) is left for future work.
- **Single retriever.** All experiments use FAISS with `all-MiniLM-L6-v2`. Different retriever architectures or similarity functions may change retrieved passage distributions and affect results.
- **NLI scorer assumptions.** The DeBERTa-v3 NLI model is imperfectly calibrated for short factoid answers and may misclassify entailment in edge cases.

9 Conclusion

We introduced a counterfactual perturbation benchmark for measuring how RAG pipeline generators respond to controlled retrieval failure. By holding retrieval architecture constant and varying only context content across four conditions (original, paraphrase, entity swap, and negation), we isolate generator behavior from retrieval quality and measure graceful degradation directly.

Faithfulness drops significantly on corrupted conditions ($p \leq .01$ in all cases) and abstention rises with medium-to-large effect sizes ($h = 0.34$–0.54). Hallucination rates remain low and largely non-significant, identifying over-abstention rather than confabulation as the main failure mode. Qwen2.5:7b and Mistral:7b achieve the strongest abstention cliffs; the Llama family shows flatter, less condition-sensitive responses. Scale does not reliably predict calibration: the 8B Llama model scores lower than both 7B competitors on Δ_{abstain}.

We release the benchmark code, dataset, and evaluation scripts as an open-source, locally-runnable toolkit requiring no API keys or cloud infrastructure. Future directions include scaling to $1{,}000+$ examples, extending to additional QA datasets (TriviaQA, HotpotQA), evaluating closed-source models, adding human quality validation of perturbation outputs, and exploring contrastive fine-tuning for improved abstention calibration.

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A Prompt Templates

Generator prompt.

System: You are a precise question-answering assistant. Answer the question using ONLY the information provided in the context. If the context does not contain enough information to answer the question, respond with exactly: I don't know. Do not add information from your own knowledge. Keep your answer concise — one sentence or less.

User:

```
Context:
[1] {retrieved_context}

Question: {question}

Answer:
```

Paraphrase generation prompt.

System: You are a precise paraphrasing assistant. Rewrite the given sentence using different wording, but preserve every factual detail exactly — names, dates, numbers, and all other specifics must remain correct. Return only the rewritten sentence with no extra commentary.

User: {`gold_context`}

B Error Analysis Breakdown

Table 7 reports the full error-analysis category counts across all four models and both corrupted perturbation types ($N = 150$ per cell, 4 models = 600 records per perturbation type).

Table 7: Error analysis category counts (all four models combined, corrupted conditions only). Cat. (a): correct despite corrupted context (parametric memory). Cat. (b): followed corrupted context (context-follower, faithfulness > 0.70, EM false). Cat. (c): abstained on corrupted condition. "Other": wrong answer, low faithfulness, not abstained.

Category	**Entity swap**	**Negation**	**Total**
(a) Parametric memory (correct)	40	66	106
(b) Context-follower (wrong)	164	135	299
(c) Appropriate abstention	272	302	574
Other (wrong, low faith)	124	97	221
Total	600	600	1200

Category (b) is the most actionable failure mode: 299 instances where the model trusted and repeated incorrect context with high faithfulness. Entity swap triggers more context-following than negation (164 vs. 135), consistent with the finding that plausible wrong entities are harder to detect than explicit logical contradictions. Category (c) dominates numerically and represents correct behavior on corrupted conditions; the concern raised in Section 5.4 is that these same models also abstain at high rates on clean original context.